

## UNDERSTANDING THE IMPORTANCE OF STRENGTH



## WHAT IS STRENGTH?

Strength is the 'ability to overcome or counteract external resistance by muscular effort' (Zatsiorsky, 1995). And in the words of Tony Yaniro (the first person to climb F8a): 'If you can't do the moves, then there's nothing to endure.'

Strength manifests in countless (and at times incomprehensible and unrealistic) ways, and it can be hard to tell what type of strength is involved in any given action. In truth, it all comes down to a single type of strength: maximum strength.

According to the *Oxford English Dictionary*, 'maximum' is defined as: 'the greatest amount, extent or intensity possible, permitted or recorded'. Maximum strength sports are usually thought of as powerlifting and weightlifting, which have nothing to do with climbing. But this doesn't mean that max strength isn't used in other sports. According to Balsalobre-Fernández and Jiménez-Reyes (2014), in reality, anyone in any sport will try to use as much strength as possible, as often as possible and as fast as possible. To understand this, González-Badillo (2002) gives two examples. Firstly, a marathon isn't won by the runner with the best endurance but by the fastest runner, as they can move their body more quickly, exert greater force with each step and therefore produce more speed. Secondly, a 100-metre sprint is won by the runner who has spent the least amount of time exerting force, the least amount of time in contact with the ground and who has needed the least amount of time to propel themselves towards the finish line. This would suggest that any given movement in sport is not just about (maximum) strength, but the crucial link between max strength and speed.

Based on this, we could say any one person has as many different measures of max strength as there are movements or actions in their sport. Maximum strength can therefore be defined as the maximum amount of strength that can be applied by an individual in response to a specific stress or load within a specific movement in sport (Balsalobre-Fernández and Jiménez-Reyes, 2014).

However, maximum strength isn't only connected to speed, it's also closely connected to endurance. This is based on the idea that increasing our maximum strength also increases our submaximal strength. For example, imagine someone who can squat a maximum of 100 kilograms in a single rep (known as 1RM or one-repetition maximum). If they then squat 80 kilograms, they'll be using 80 per cent of their max strength. If they subsequently increase their 1RM to 160 kilograms, squatting 80 kilograms becomes 50 per cent of their max strength instead of 80 per cent. In other words, it's the same workload but less effort, meaning they can do more reps at this weight.

Let's apply this to climbing. If a climber struggles to do one pull-up, each time they need to pull when climbing they'll use a very high percentage of their strength, or even their max strength, and they'll fatigue quickly. If this climber could do a pull-up with say 50 kilograms of added weight, then a pull-up with no added weight wouldn't be much effort at all and they'd be able to do lots of them. As a result, pulling when climbing wouldn't be nearly so tiring.

This clearly shows that greater max strength equals greater endurance – and why strength is the mother of all physical qualities!

Endurance could also be defined as the ability to keep producing the necessary force to move mass through space over a set period of time. In other words, the ability to produce enough force to move your body at 13 miles per hour if you want to run a two-hour marathon, or to climb a 40-metre overhang without resting on the rope.

## TYPES OF MUSCLE CONTRACTION

How do muscles contract to generate force? There are three different types of muscle contraction:

- *Isometric*: the muscle comes under tension but remains the same length. This happens when we push against an unmovable object (like a wall) or when we hold a certain body position (like a lock-off). In climbing, it's the main type of forearm muscle contraction: when we grip a hold, our forearm flexor muscles contract but remain the same length as we move the rest of our body.
- *Isotonic*: the muscle comes under tension and the length of the muscle changes. Concentric isotonic contractions are when the points of origin and insertion move closer together and enough force is generated to move an object (for example, the upward phase of a pull-up). Eccentric isotonic contractions are when the points of origin and insertion move further apart as sufficient force cannot be maintained or we want to return to the initial position (the downward phase of a pull-up).
- *Auxotonic*: this is a mixture of all the above, which is what happens in reality in all sports. With auxotonic contractions, a muscle could start off under isometric tension, instantly switch to concentric and then back to isometric or to eccentric, depending on how the action or movement evolves. Muscle contractions like to mix things up a little!

## TYPES OF MUSCLE FIBRE AND RECRUITMENT

Sticking with types ... Each muscle contains different types of fibres. The body decides which fibres to contract or recruit depending on the action that we want to perform.

- *Type I fibres*: also known as slow-twitch, aerobic or red fibres. These muscle fibres are used for actions like walking or jogging that don't require a sudden burst of energy. They mainly use the aerobic energy system, and they are red because they contain red blood cells. As such, they're highly vascularised with a large number of capillaries for the supply of blood and nutrients, and they are highly resistant to fatigue.
- *Type II fibres*: also known as fast-twitch, anaerobic or white fibres. These include the subtypes IIa, IIb and IIx. Basically, IIb fibres have a higher recruitment threshold than IIa fibres, meaning they're only activated in more intense muscle contractions, and IIx fibres are indeterminate (they're transformed into IIa or IIb fibres as needed). Type II fibres mainly use the anaerobic energy system and are white because they don't need the oxygen that is carried in red blood cells to produce energy. Although they can produce lots of energy, they fatigue much more quickly than type I fibres.

The percentage of type II fibres in our muscles is determined by our genetics. Through training, we can change type II muscle fibres into type I, but there's no scientific evidence that we can change type I into type II. This can be problematic in sports that require both strength and endurance, such as mountaineering. A mountaineer needs lots of type I fibres to get them through gruelling approaches, long routes and descents, but they also need type II fibres to do technical and difficult moves while carrying all their gear.

To adjust the percentage of each type of muscle fibre to best suit our needs, we need to follow an effective and well-designed training plan. We'll look at this in more depth after piecing together the other parts of the puzzle.

## CAUSES OF FATIGUE

Why do we run out of steam? Why does our strength fail us? Why can't we maintain our energy levels? What turns off the supply? Firstly, it's important to distinguish between the type of fatigue caused by an 800-metre route, after a four-hour approach with 1,500 metres of elevation, and the type of fatigue caused by a six-move boulder problem. Secondly, it's important to distinguish between general or central fatigue and peripheral or local fatigue. Even then, it's not that simple, and in reality there are many factors that can contribute to fatigue (Cairns, 2013):

- *Build-up of lactate*: although in the past lactate has been blamed for causing fatigue, recent studies suggest that neither intracellular nor extracellular lactate reduces performance.
- *Metabolic acidosis*: at body temperature, acidosis reduces the ability to generate force, it significantly reduces the speed of muscle contractions and it causes a slight loss of calcium sensitivity on a cellular level.
- *Muscle hypoxia*: high-intensity muscle contractions need a good supply of oxygen to be sustained over time. Hypoxia refers to a lack of oxygen, where the muscle runs out of oxygen and can't draw any energy from its fuel supply. The muscle then competes with the brain for oxygen, lowering the brain's oxygen supply and leading to central fatigue.
- *Phosphates*: high-intensity exercise depletes adenosine triphosphate (ATP) and creatine phosphate (PCr) energy reserves, increasing the presence of inorganic phosphates. This affects the function of contractile proteins, inhibiting the production of force and speed.
- *Carbohydrate availability*: the availability of muscle and liver glycogen stores is a key factor in central fatigue but has little effect on local fatigue.
- *Build-up of hydrogen ions ( $H^+$ )*: there's strong scientific evidence that the acidosis caused by a build-up of hydrogen ions significantly inhibits force, power and shortening velocity in muscle cells.
- *Build-up of ammonium*: the production of ammonium has also been linked to fatigue. Ammonium is produced when concentrations of ATP/PCr fall below critical levels and branched-chain amino acids are catabolised. This occurs in high-intensity training (for example, strength training, sprints, powerful boulder problems). In these cases, ammonium build-up is partly responsible for and an indicator of local muscle fatigue, and it can also contribute to central fatigue (Balsalobre-Fernández and Jiménez-Reyes, 2014).

However, between a movement command leaving the cerebral cortex, travelling through the motor neurons in the spinal cord, entering the muscle cell and arriving at the actin and myosin muscle fibres, there are many different places where fatigue could appear. As such, in reality we should be thinking about neuromuscular fatigue.

## MUSCLE FAILURE: IS IT REALLY NECESSARY?

The 'no pain, no gain' mantra, training until you can't do another rep or move ... Is training to failure really necessary? Before we decide, let's look at some scientific evidence:

- Training to failure causes excessive fatigue, the stimulus creates muscle fibre transformation (IIX to IIa) and lowers the rate of force development (RFD) (Andersen and Aagaard, 2010; Marshall *et al.*, 2011; Schuenke *et al.*, 2012; Pareja-Blanco *et al.*, 2017).
- Training at half the possible number of reps leads to greater intra-set speed, lower performance loss and better neuromuscular recovery post-training. It results in less hormonal stress (cortisol, prolactin, GH, IGF-1), less muscle damage (creatine kinase (CK)) and a lower reduction in heart rate variability and complexity (HRV/HRC) post-training (González-Badillo *et al.*, 2016).
- Pareja-Blanco *et al.* (2017) compared the strength gains in two groups tasked with training squats: one group trained at half the possible number of reps, at 20 per cent velocity loss (VL20), while the other group trained at 40 per cent velocity loss (VL40), almost to muscle failure. Both groups made the same strength gains, but VL20 showed greater improvement in the countermovement jump test and VL40 showed greater hypertrophy with more type IIX to IIa muscle-fibre transformation.

So, what do we think? To build strength, it seems obvious that we should steer well clear of muscle failure, high levels of fatigue and the 'no pain, no gain' approach. (Don't worry if you don't know what velocity loss or RFD is, as we'll look at these later on in [chapter 7](#).)

Training for any sport (with the exception of bodybuilding and very few others) should focus on the maximum performance of each rep, set and exercise. If we train to failure or exhaustion, fatigue will significantly limit the intensity at which we're able to perform these actions: we won't be able to perform at maximum intensity because we're tired. This doesn't just apply to intra-session fatigue: intra-microcycle fatigue, day-to-day fatigue and inter-session fatigue also have a significant effect. We should train in a way that creates the least possible fatigue so that

we can come back and give it our all in our next session. As Yoda says: 'Do ... or do not. There is no try.' It's all or nothing: maximum intensity or it's not worth it. And you can't do this when you're tired.

## WAYS TO DEVELOP STRENGTH: HYPERTROPHY AND NEURAL

We can all agree that Adam Ondra and Arnold Schwarzenegger are both really strong, but they couldn't look any less alike. This is because they use different training methods.

One of the aims of this book is to debunk the idea that you should only train strength if you want to look like Arnie. Traditionally, we've viewed bodybuilding practices as the only type of strength training and physical conditioning, leading to unwanted weight gain and perpetuating the myth that weight training makes you slow and heavy. Fortunately, this is gradually starting to change.

There are a few different options for making our muscles stronger:

- *Hypertrophy (sarcoplasmic/aesthetic or myofibrillar/functional)*: enhancing contractile capacity by increasing the size of muscle fibres and contractile proteins (actin and myosin). Larger muscles ... just like Arnie.
- *Hyperplasia*: increasing the number of muscle fibres instead of the size. However, there's no evidence that it's possible to increase the number of muscle fibres in human beings, so this isn't an option.
- *Neural hypertrophy*: the best option for climbers. The following analogy can explain this type of training. A Ferrari won't use its full horsepower to drive around a city; it'll save maximum power for overtaking or racing. Our muscles are similar: they have a set number of muscle fibres (horsepower) and will use more or less of them as needed. The difference, and the problem, is that when some of these fibres aren't fired up or recruited, they're still there but we can't use them. This is where neural hypertrophy comes into play. It increases the *pool* (quantity) of fibres that are activated or innervated by a single motor neuron, aiming to maximise the innervation ratio and make use of every last gram of muscle. Greater fibrillar recruitment requires a more intense signal to be transmitted from motor neurons to muscle fibres. This electrical impulse is known as the action potential. We can't make our nervous system transmit a stronger action potential to the muscle fibres, but we can make it send more action potentials per unit of time. This is how we can recruit new fibres and get stronger without getting heavier.

The following table shows the key variables of strength training (adapted from Benito, 2008):

| Rate of protein degradation (INTENSITY) | Mechanical workload (REPETITIONS) | Total amount of degraded proteins (HYPERTROPHY) |
|-----------------------------------------|-----------------------------------|-------------------------------------------------|
| Low (<60% 1RM)                          | High (15+)                        | Low                                             |
| Moderate (60–85% 1RM)                   | Moderate (6–12)                   | High                                            |
| High (85–100% 1RM)                      | Low (1–4)                         | Low                                             |

As the table shows, both a high number of reps with lighter weights and a low number of reps with heavier weights will achieve a low degree of hypertrophy. And since the aim of strength training for climbing should be to improve our strength-to-weight ratio, we should be training as per the last row of the table: heavy weight and low reps.

## STRENGTH TRAINING FOR INJURY PREVENTION

Is strength training also useful for injury prevention? Absolutely. Let's use another car analogy. Imagine driving a 70-horsepower car over a mountain pass. You'll be in a very low gear, going very slowly, burning a load of fuel and pushing the revs into the red. If you do this often or you want to go faster than the car can handle, you'll risk blowing the engine. Then think back to our Ferrari: with 500 horsepower, it will cruise over the pass at 60 miles per hour and at barely 2,000 revs per minute. Basically, a lot faster and with a lot less effort: it's unlikely you'll blow the engine. The same applies with the human body and the idea behind maximum strength training: the stronger you are, the less relative effort will be required for the same workload. Imagine a weaker and a stronger climber trying a hard move: the weaker climber is less likely to do the move, they'll fatigue faster and to a higher degree, will inflict greater damage on their muscles and tendons, and will be more likely to injure themselves.

Let's take a look at some scientific research:

- A study performed by Lauersen *et al.* (2014) analysed 26,610 subjects and a total of 3,464 injuries. It concluded that an appropriate strength programme was the most effective way to reduce injury, leading to a 30 per cent reduction in the number of injuries and a 50 per cent reduction in injuries related to overloading. Multicomponent and proprioception training programmes had a moderate effect but were significantly less effective than strength training. In turn, stretching proved to have no effect whatsoever on reducing injury.
- A study performed by Malone *et al.* (2019) explored the link between training load, physical condition and injury in amateur team sports (two to three training sessions a week). Participants who were relatively stronger (strength-to-weight ratio), faster over 5-metre, 10-metre and 20-metre sprints, and able to do more sprints in a short period of time showed greater tolerance of higher training loads and were less likely to injure themselves than participants with less well-developed physical qualities.

These findings suggest a new paradigm that many people are still unaware of: strength training is a more effective means of injury prevention than stretching. If this is news to you, steady yourself for the findings of Jenkins *et al.* (2016). They concluded that higher training loads (80 per cent 1RM) lead to greater strength gains as they produce greater neural adaptation than lower training loads (30 per cent 1RM). So, training heavy is even more effective than training light and much more effective than stretching for injury prevention. In short, it is our focus on strength training that will help minimise the risk of injury. We can do this in two ways:

- *Training to maximise climbing performance:* if a climber can hang an 8-millimetre edge, using a 22-millimetre edge will cause minimal stress on their muscles and tendons. However, for novices or climbers at lower grades, using a 22-millimetre edge could be a huge challenge (even with feet on) for both the climber and their tendons. So, by training to maximise performance, we'll be less likely to injure ourselves.
- *Training to prevent/correct strength imbalance:* like any other sport, climbing involves the repetition of a specific set of movements or actions (for example, gripping holds, pulling up, pulling over). This means that the muscles used to perform these movements (agonists) become stronger. As these muscles increase in strength, they exert an increasingly strong pull on the corresponding joint. If the muscles that stabilise these joints (antagonists) can't respond with sufficient opposing force, this imbalance will result in joint instability. In turn, this can lead to poor posture, contracture or spasm in the antagonist muscles as they attempt to stabilise the joint by resisting the pull of the agonist muscles, and a significantly higher risk of injury.